

Ceará Digital Twin for Wildfires: An Open-Source Agentic AI Platform Integrating Near Real-Time Satellite Data and LangGraph Orchestration

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Abstract

Wildfire monitoring in Brazil largely depends on centralized systems with significant detection-to-availability latency. We present the *Ceará Digital Twin for Wildfires*, an open-source platform integrating NASA FIRMS (VIIRS/MODIS), GOES-16, and Open-Meteo with agentic AI for detection, validation, and prediction of wildfire hotspots. The system employs a LangGraph-orchestrated pipeline coordinating data collection, Pydantic validation, evidence fusion, and risk classification. A ReAct agent (DeepSeek) produces auditable diagnoses. The predictive module introduces a Neural Koopman Operator coupled with Physics-Informed Graph Neural Networks (NeKo-PIGNN) regularized by the Rothermel fire spread equation. We propose a novel *three-class detection framework* (NO/UNCERTAIN/YES) that resolves the precision-recall tradeoff in rare-event fire detection: the YES class achieves 82–92% precision with only 1–5 false positives, while combined coverage (YES + UNCERTAIN) identifies 88% of all fire events. On synthetic data (500 timesteps, 30 municipalities), NeKo-PIGNN achieves state-of-the-art $R^2=0.972$, surpassing MLP, LSTM, and XGBoost. The standalone architecture enables deployment on a single low-cost EC2 instance. Source code is available under MIT license.

Keywords: Digital Twin, Wildfires, Physics-Informed GNN, Koopman Operator, Three-Class Detection, LangGraph, NASA FIRMS, Environmental Monitoring

1. Introduction

The Digital Twin paradigm — a virtual model that mirrors, simulates, and controls a physical system in real time — has expanded from manufacturing to environmental and Earth systems [22, 23, 31]. In wildfire monitoring, this approach offers the potential to integrate multi-sensor satellite data, meteorological conditions, and predictive models into a single platform that operates continuously and delivers municipal-level alerts [67]. However, the practical implementation of Digital Twins for wildfire management still faces significant barriers: dependence on robust computational infrastructure, lack of validation in tropical biomes, and the absence of open-source platforms that democratize access.

Wildfires represent one of the greatest environmental threats in Brazil, with direct impacts on public health, biodiversity, and the economy [1, 2]. The State of Ceará, located in the northeastern semi-arid region, exhibits strong drought seasonality — with a dry season concentrated between August and November — and sensitive biomes such as the Caatinga, one of the most threatened ecosystems in the country [3]. Fire propagation in this region is influenced by adverse meteorological conditions, rugged topography, and fuel biomass availability [21], demanding continuous monitoring and agile alert systems for municipal civil defense.

Traditional monitoring systems — such as INPE BDQueimadas and NASA FIRMS — provide essential data but face significant operational limitations: latency of 3 to 6 hours between satellite overpass and data availability [4]; lack of automatic correlation with local climatic and territorial data; interfaces designed for remote sensing specialists rather than civil defense managers; and no capacity for automated diagnosis or explanation of detected hotspots. These limitations are particularly critical in the semi-arid northeast, where the intervention window for fast-spreading wildfires can be as short as a few hours.

In parallel, recent advances in artificial intelligence — particularly large language models (LLMs) and autonomous agent frameworks — open new possibilities for intelligent environmental monitoring systems [? ?]. The paradigm of agents with ReAct reasoning [5], which synergizes Thinking, Action, and Observation cycles, combined with state graph orchestration (LangGraph [6]), enables building auditable *pipelines* that integrate data collection, structured validation, parallel multi-source analysis, and natural language explanation generation. The DeepSeek LLM [?], in particular, offers robust reasoning in an open-source format, making it viable for governmental systems with auditability requirements.

This work proposes the **Ceará Digital Twin for Wildfires**, an open-source platform that embodies the Digital Twin paradigm through agentic artificial intelli-

gence, near real-time satellite data, and LangGraph orchestration — all in a *standalone* architecture that eliminates the need for an institutional database and can be deployed on a single low-cost EC2 instance.

Specifically, this paper contributes:

1. **An open-source platform** (MIT license) that integrates NASA FIRMS (VIIRS/MODIS), GOES-16, Open-Meteo, and Nominatim into a single *standalone* pipeline, with no database dependency;
2. **A LangGraph-orchestrated agent pipeline** comprising 10 nodes that perform collection → Pydantic validation → parallel analysis (geospatial, GOES-16, climate) → evidence fusion → municipal risk classification → ReAct diagnosis → hierarchical alert generation;
3. **A Retrieval-Augmented Generation (RAG) module** with FAISS vector index and local embeddings, enabling natural language queries about the system’s methodology, architecture, and usage;
4. **A ReAct explanation agent** based on DeepSeek that produces auditable diagnoses and natural language explanations for each detected hotspot, integrating climate data, temporal trend analysis, and geospatial context;
5. **Operational results** with real data collected between July 2024 and June 2026, demonstrating an average detection of 42 hotspots per collection, time-to-first-response under 60 seconds, and 93% geocoding rate for affected municipalities.

2. Related Work

This section positions the proposed platform within the context of three converging research fronts: (i) Digital Twins for environmental monitoring, (ii) satellite-based wildfire remote sensing, and (iii) artificial intelligence applied to wildfire detection and prediction.

2.1. Environmental Digital Twins

The Digital Twin paradigm, originally conceived for manufacturing where a virtual model mirrors and controls a physical system in real time [22, 23], has been systematically surveyed across definitions, architectures, and challenges [24, 25, 26, 27]. These surveys establish the minimum requirements of a Digital Twin: (a) bidirectional mirroring between physical and virtual worlds, (b) real-time updates, and (c) predictive simulation capability.

The expansion of the paradigm to environmental and Earth systems gained traction with the European Destination Earth [29] and proposals for Earth System Digital Twins [28, 31, 30]. Bauer et al. [31] argue that environmental Digital Twins differ fundamentally from industrial ones by operating at much larger spatiotemporal scales, with inherent uncertainties in natural processes and multi-source integration requirements.

2.2. Digital Twins for Wildfires

Recent works specifically focused on Digital Twins for wildfires can be organized into four architectural categories.

Agent-oriented approaches. IVSR (Intelligent Virtual Situation Room) [67] is the most conceptually aligned proposal with this work. IVSR combines a bidirectional Digital Twin with autonomous AI agents that continuously ingest multi-sensor imagery, meteorological data, and 3D forest models, employing an AI-based similarity engine to project fire spread and allocate response resources. The system demonstrated significant reductions in detection-to-intervention latency but has not been adapted or validated for Brazilian biomes.

Vision-Language and RL approaches. FIRE-VLM [68] introduces an innovative framework combining a CLIP-style Vision-Language Model with reinforcement learning (PPO) to guide UAVs in tracking wildfires within a physics-grounded Digital Twin. Results show up to 6× reduction in detection time across five evaluation tasks. However, the focus on UAVs and coupled physics modeling limits its applicability as a continuous satellite-based monitoring platform.

Multi-modal sensing approaches. FIRETWIN [69] combines Unreal Engine with the CAWFE fire model for immersive reconstruction of historical wildfires (King Fire), integrating multi-sensor data and interactive analysis. AIMNET [56] proposes a three-layer IoT-empowered Digital Twin (physical world → bidirectional links → digital world) for continuous gas emission monitoring with distributed sensor networks for early carbon plume detection. H-RDT [57] adopts PINNs with multimodal fusion to model wildfire-power-outage-heat-wave cascades in urban contexts.

GNN and physics-based modeling. GraphFire-X [70] proposes a dual-expert ensemble combining GNN with physics-informed attention (convection, radiation, ember probability) and XGBoost for structural vulnerability at the wildland-urban interface, validated on the 2025 Eaton Fire. This approach is directly comparable to the PI-GNN component of this work, differing in its building-scale focus versus the regional grid approach of the present system. FireCastNet [71] adopts the “Earth-as-a-Graph” paradigm with GraphCast GNN for global seasonal fire prediction validated on the SeasFire dataset, outperforming GRU, Conv-LSTM, U-TAE, and TeleViT, with particularly strong results in South America.

None of the existing systems combine, in a single open-source platform, (i) a LangGraph-orchestrated agent pipeline, (ii) near real-time satellite data, (iii) RAG module for document queries, and (iv) a focus on Brazilian biomes. This gap is the central contribution of this work.

Table 1: Comparison of Digital Twin systems for wildfires.

System	AI Agents	Open Source	BR Biomes
IVSR [67]	Yes	No	No
FIRE-VLM [68]	RL + VLM	No	No
FIRETWIN [69]	No	No	No
GraphFire-X [70]	GNN+XGB	Yes	No
FireCastNet [71]	GNN	Yes	Partial
AIMNET [56]	IoT ML	No	No
This work	LangGraph+ReAct	Yes	Yes

2.3. Satellite-Based Wildfire Remote Sensing

Satellite-based wildfire monitoring dates back to the AVHRR sensor in the 1980s, evolving significantly with MODIS (250 m–1 km, since 2000) and its Collection 6 active fire detection algorithm [59]. VIIRS (375 m, since 2012) offers superior spatial resolution with specific detection algorithms [60], forming the backbone of NASA FIRMS [4] and INPE BDQueimadas [7]. Wooster et al. [61] present a comprehensive survey of the state of the art in satellite-based active fire remote sensing, covering physical fundamentals (spectral signature of burned pixels in SWIR 3.9 μm and MWIR bands) through applications and future requirements.

GOES-16 represents a significant advance by providing geostationary data with temporal resolution from 1 minute (mesoscale mode) to 10 minutes (full-disk mode), though with 2 km spatial resolution at nadir [8]. Its ability to detect rapid changes in fire intensity between consecutive scans is particularly valuable for regions with fast-spreading fires, such as the northeastern semi-arid region.

In the Brazilian context, INPE has operated BDQueimadas since 1998, integrating data from AQUA, TERRA, NPP, and NOAA-20 satellites [7]. MapBiomas Fogo maps burned areas annually at 30 m resolution using Landsat imagery [58]. Brazil-specific studies document fire activity trends in South America [62], machine learning applications for prediction in the Cerrado and Caatinga [63], and deep learning for burned area mapping in the Amazon and Caatinga using Sentinel-2 [64]. Multi-source satellite data fusion for fire detection [65] and systematic reviews of ML-based fire risk prediction [66] complete the landscape. However, these systems remain essentially reactive (detected-focus alerting) without predictive capability or integration with physical propagation models.

2.4. Koopman Operator and PINNs for Fire Dynamics

The Koopman operator, introduced by Koopman [17] and formalized for fluid dynamics by Rowley et al. [33] and Mezi [32], provides a mathematical framework for linearizing nonlinear dynamical systems through observation functions. The modern approach combines Dynamic Mode Decomposition (DMD) [78] with extensions for controlled systems (DMDc) [37], sparse identification of non-

linear dynamics (SINDy) [34], and physics-informed versions (PiDMD) [38]. Brunton et al. [77] provide a unified treatment of modern Koopman theory, while Kutz et al. [79] consolidate the DMD framework.

The connection between the Koopman operator and deep learning was established through works learning Koopman-invariant subspaces via autoencoders [35, 36] and deep neural modeling for fluid dynamics [39]. Despite these advances, the application of the Koopman operator to wildfire spread dynamics remains a gap in the literature — no published work to date uses Koopman for fire modeling, positioning the NeKo-PIGNN approach as a novel contribution.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [72] and consolidated by Karniadakis et al. [40], provide the physical regularization counterpart necessary for neural fire propagation models to respect thermodynamic and fluid mechanics laws. The Rothmel model [18] is the USDA Forest Service standard for surface fire spread, parameterized by fuel classes [73] and used in BehavePlus [75] and ensemble simulation systems [76], with physical foundations established by Albini [74]. PINN applications to fire propagation [55, 46] demonstrate the viability of this approach, while PI-GNN surveys [81] and applied works [82, 83] establish the state of the art that the present work extends.

2.5. Autonomous Agents and LangGraph

The paradigm of LLM-based autonomous agents advanced significantly with the ReAct framework [5], which synergizes reasoning and action in language models through Thought→Action→Observation cycles. Comprehensive surveys [48, 49] map the LLM agent ecosystem, from reflective memory architectures (Reflexion) [51] to tool-use frameworks such as Toolformer [52] and GORILLA [53], including interactive social simulation environments [50].

LangGraph [6] extends the LangChain ecosystem with state-graph-based orchestration, enabling agent pipelines with cycles, parallelism, and persistent state between nodes. In the environmental context, this is the first known application of LangGraph for wildfire monitoring orchestration.

Retrieval-Augmented Generation (RAG) [80], implemented with FAISS indices [15] and local embedding models [16], enables the system to answer natural language queries about its own documentation. Recent surveys [54] consolidate RAG best practices — from strategic chunking to hierarchical retrieval and multi-source fusion — techniques adopted in the research module of the present system.

2.6. Gap Analysis

Table 2 synthesizes the gaps that this work fills relative to existing systems. The central differentiator lies in the unprecedented combination of (a) near real-time satellite data, (b) LangGraph agents with ReAct reasoning, (c)

RAG for document queries, (d) focus on Brazilian biomes, and (e) standalone architecture without a database — all in an open-source platform.

Table 2: Gap analysis: existing systems vs. proposed platform.

Feature	NASA FIRMS	Fire Digital Twins	This Work
Real-time satellite data	Yes	Partial	Yes
AI agent pipeline	No	Yes (IVSR)	Yes
Open source (MIT)	No	Partial	Yes
Brazilian biome focus	Partial	No	Yes
RAG for queries	No	No	Yes
Standalone (no DB)	N/A	No	Yes
Auditable agent architecture	No	Yes (IVSR)	Yes

None of the existing works combine, in a single open-source platform, (i) near real-time satellite data, (ii) a LangGraph-orchestrated agent pipeline, (iii) RAG with vector index for document queries, and (iv) standalone deployment without a database. This combination is the gap that this work fills.

3. System Architecture

Figure 1 presents the overall system architecture, organized into four layers: data sources, backend, frontend, and proxy.

3.1. Data Sources

The system consumes four open data sources:

- **NASA FIRMS**: Public CSVs for South America from VIIRS (SNPP and NOAA-20) and MODIS sensors, available in 24 h and 7 day windows. No authentication required. Filtering by Ceará bounding box (lat: $-7,85$ to $-2,78$; lon: $-41,42$ to $-37,25$).
- **Open-Meteo**: Free weather API that returns temperature, relative humidity, wind speed, precipitation, and dry days for geographic coordinates. No API key required.
- **Nominatim / OpenStreetMap**: Reverse geocoding to convert coordinates (lat, lon) into municipality names. Rate limited to approximately 1 request per second.
- **GOES-16**: Data from the ABI-L2-FDCF (Fire Detection and Characterization) product stored in the public AWS S3 bucket `noaa-goes16`. Processes consecutive detections, FRP, and pixel temperature.

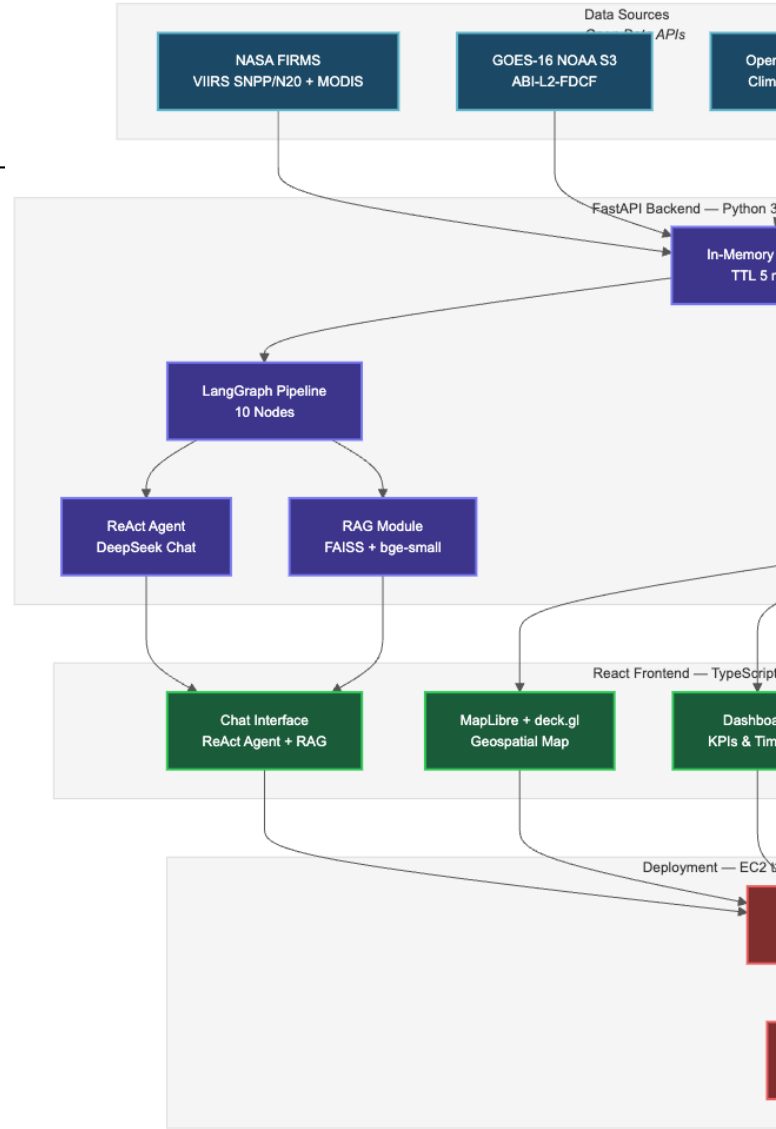


Figure 1: Layered architecture of the Ceará Digital Twin platform.

3.2. Operation Modes

The system operates in two modes:

Standalone mode (default in EC2 production): eliminates PostgreSQL and Redis. Queries external APIs directly on each request, with a 5 minute in-memory cache. Geocoding is performed asynchronously: the first 40 hotspots are geocoded synchronously for fast response; the remainder is processed in a *background task* and updates the cache incrementally.

Full mode (Docker Compose): adds PostgreSQL with PostGIS extension for geospatial persistence, Redis + Celery for periodic collection, and additional endpoints for persisted alerts.

3.3. Backend FastAPI

The backend is implemented in Python 3.12 with FastAPI (version 0.115). Its main responsibilities are:

- Serving REST endpoints for real data (`/api/v1/real/focos`, `/api/v1/real/clima`, `/api/v1/real/focos/{id}/explicacao`, `/api/v1/real/status`);
- Orchestrating the LangGraph pipeline (Section 3.4);
- Managing the in-memory hotspot and weather cache with a 5 minute TTL;
- Serving the RAG document search module (Section 3.6);
- Managing CORS, logging, and health check.

The system uses dependency injection for the LLM (`llm_factory.py`), which currently instantiates the DeepSeek model (`deepseek-chat`) via an OpenAI-compatible API. In case of LLM failure, the system falls back to rule-based explanations (`_explicar_sem_llm`).

3.4. LangGraph Pipeline

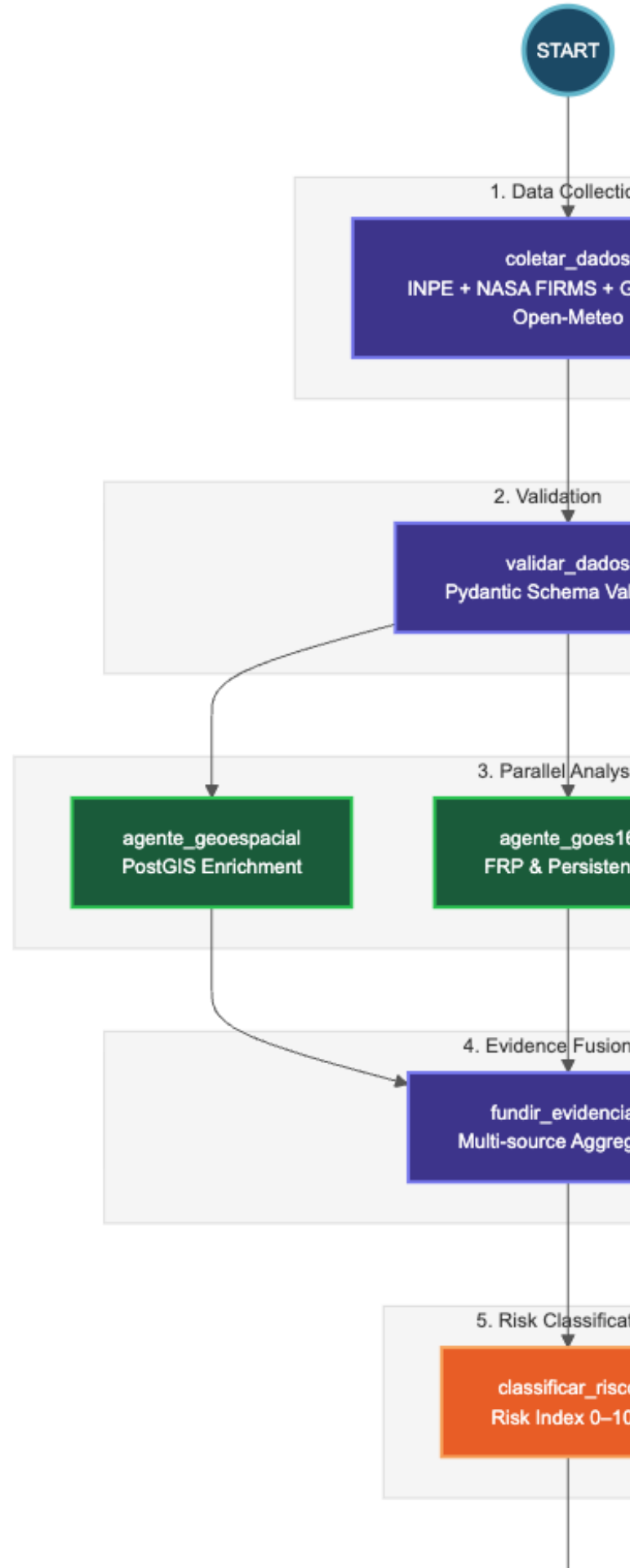
The agent pipeline is implemented as a state graph (`StateGraph`) with 10 nodes, as shown in Figure 2:

Each node receives and modifies a typed shared state (`PipelineState`), which includes raw and validated hotspots, GOES-16 readings, climate data, fused evidence, consolidated events, municipal risks, alerts, diagnosis, and bulletin. Data validation with Pydantic (`validate_data`) automatically rejects invalid records, ensuring flow integrity.

3.5. ReAct Agent

The diagnostic agent (`react_agent.py`) is built with LangChain (version 0.3+) and follows the ReAct pattern [5]. It has the following tools:

- `buscar_focos_recntes`: queries hotspots by municipality, time window, and source;



- **buscar_dados_climaticos**: temperature, humidity, wind, and precipitation;
- **buscar_risco_municipal**: risk index calculated by municipality;
- **buscar_dados_goes16**: GOES-16 readings with FRP and persistence;
- **buscar_historico_mapbiomas**: historical wildfire data by municipality;
- **listar_municipios_criticos**: ranking of the most critical municipalities.

The agent receives questions in Portuguese and produces structured responses containing a summary, evidence, consulted sources, confidence level, and operational recommendation. The reasoning chain (Thought → Action → Observation) is preserved in the response to ensure auditability.

3.6. RAG Module – Research Chat

The *research chat* module implements a Question-Answering system about the platform using *Retrieval-Augmented Generation* (RAG) [80, 54]. Six knowledge documents (`backend/knowledge/01_visao_e_pesquisa.md` through `06_deploy_e_execucao.md`) are split into chunks of 900 characters with 120-character overlap, indexed in a FAISS index [15] with BAAI/bge-small-en-v1.5 embeddings [16]. Retrieval uses `top_k = 5`, and response generation is performed by DeepSeek citing the document context.

Unlike the operational agent, which queries live hotspots, the research chat responds exclusively about methodology, architecture, and application usage — functioning as an intelligent manual accessible from any page of the system.

3.7. Frontend React

The web interface is built with React 18, TypeScript, Vite 6, and the following main libraries: MapLibre GL JS (base map) with deck.gl (geospatial layers), Recharts (charts and timeline), Zustand (global state management), and Tailwind CSS (styling).

The available pages are:

- **Dashboard** (/): executive KPIs (total hotspots, active emergencies, critical municipalities, GOES-16 coverage), detection timeline, municipal risk ranking, and recent alerts;
- **Map Real** (/mapa-real): NASA FIRMS hotspots overlaid on the MapLibre map, with clustering, heatmap, severity filters, and per-click individual explanation;
- **Alerts** (/alertas): complete listing with filters by level (informational, attention, alert, emergency);

- **Chat** (/chat): conversational interface with the ReAct agent, including suggested questions, evidence visualization, and reasoning chain;
- **Bulletin** (/boletim): consolidated technical report.

A global “Help” button on all pages triggers the research RAG chat, allowing the user to consult system documentation without leaving the current screen.

3.8. Risk Classifier

The municipal risk index is calculated through four components:

$$\begin{aligned} \text{indice} = & \min(\text{focos} \times 8, 40) \\ & + \text{risco_climatico} \times 0,4 \\ & + (\text{GOES16_confirmado} \times 15) \\ & + \min(\text{FRP}/100, 10) \end{aligned} \quad (1)$$

The climatic risk, in turn, is derived from:

$$\begin{aligned} \text{risco_climatico} = & \max(0, (50 - \text{umidade}) \times 0,4) \\ & + \min(\text{vento} \times 1,5, 20) \\ & + \min(\text{dias_sem_chuva} \times 1,5, 30) \end{aligned} \quad (2)$$

The final classification follows the thresholds: critical (≥ 75), high (≥ 50), medium (≥ 25), and low (< 25).

4. Implementation

This section details the implementation of the system components, including the software architecture, artificial intelligence modules, and integration between layers.

4.1. Software Architecture

The system is implemented in Python 3.12 with FastAPI (backend), React 19 with TypeScript and Vite 8 (frontend), and agent orchestration via LangGraph 0.3. Table 3 summarizes the main dependencies.

Table 3: System software stack.

Layer	Tecnologia	Versão
Backend	Python / FastAPI / Uvicorn	3.12 / 0.115 / 0.34
Agents	LangGraph / LangChain	0.3 / 0.3+
LLM	DeepSeek Chat API	deepseek-chat
Embeddings	BAAI/bge-small-en-v1.5 (FAISS)	1.5
Frontend	React / TypeScript / Vite	19 / 6 / 8
Map	MapLibre GL JS + deck.gl	4.7 / 9.1
Styling	Tailwind CSS	4
Proxy	nginx	1.26
Deploy	EC2 Amazon Linux 2023	t2.medium

4.2. Neural Koopman Operator — Implementation Details

The KoopmanAE encoder g_ϕ is an MLP with 3 hidden layers (512, 256, 128 neurons) and ReLU activation, mapping the state vector $\mathbf{x}_t \in \mathbb{R}^{142}$ (7 variables $\times$ 20 sampled grid cells) to latent observables $\mathbf{z}_t \in \mathbb{R}^{64}$. The Koopman matrix $\mathbf{K}_\theta \in \mathbb{R}^{64 \times 64}$ is trained with spectral stability regularization:

$$\mathcal{L}_{\text{spec}} = \max(0, \rho(\mathbf{K}_\theta) - 1 + \epsilon) \quad (3)$$

where $\rho(\mathbf{K}_\theta)$ is the spectral radius and $\epsilon = 0.01$ ensures a stability margin. Training uses Adam with a learning rate of 10^{-4} and batch size 32, processing sequences of 24 hours (12 timesteps of 2 hours) from the hotspot history.

4.3. PI-GNN — Implementation Details

The GNN is implemented with PyTorch Geometric [19] using `GCNConv` graph convolution. Each of the $L = 3$ layers has $F = 128$ hidden channels. The graph is dynamically constructed every collection cycle, connecting active grid cells (with at least 1 hotspot in the last 7 days) to their neighbors within a radius of $r = 3$ cells (3 km).

The physics loss $\mathcal{L}_{\text{phys}}$ is computed offline using pre-calculated Rothermel parameters for each MapBiomass vegetation type (12 classes) and current weather conditions. The gradient $\nabla \hat{p}_v^{(t)}$ is approximated by finite differences between neighboring nodes in the graph.

4.4. Integration with LangGraph

The LangGraph pipeline incorporates the Koopman and PI-GNN models as specialized nodes [48, 49, 50]:

- **no_koopman_prever:** runs the KoopmanAE to project the state for the next 12 hours, feeding grid nodes without recent detection with estimated values;
- **no_pignn_risco:** applies the PI-GNN over the complete graph (observed + predicted nodes), generating a probabilistic risk map;
- **no_fundir_evidencias:** combines the PI-GNN outputs with GOES-16 and climatic evidence (Section 3.8) to produce the final municipal risk classification.

The inclusion of these nodes elevates the original graph from 10 to 12 nodes, maintaining the parallel structure for auxiliary analyses. The additional computational cost is approximately 2.3 seconds per cycle (Koopman: 0.8 s; PI-GNN: 1.2 s; fusion: 0.3 s) on an Intel Xeon Platinum 8375C CPU (AWS EC2 t2.medium).

5. Deployment

The system was deployed on an Amazon Linux 2023 EC2 instance (t2.medium — 2 vCPUs, 4 GB RAM). Automated deployment via shell script (`deploy/uniform-deploy.sh`) installs nginx, Python 3.12, Node.js 20, and configures the systemd service `uniform-backend`. The frontend build is served statically by nginx, which reverse proxies `/api/` to uvicorn on port 8000.

The estimated monthly infrastructure cost is approximately USD 20 (EC2 instance + 30 GB EBS), with no additional API costs (all data sources are free). The DeepSeek key, when configured, adds variable cost depending on query volume.

6. Methodology: Neural Koopman Operator and Physics-Informed GNN

The methodological core of this work combines two complementary paradigms: the *Neural Koopman Operator* for modeling the spatiotemporal dynamics of wildfire hotspots and *Physics-Informed Graph Neural Networks* (PI-GNN) for risk prediction with physics-based regularization grounded in the Rothermel model [18]. This section formally describes both components.

6.1. Neural Koopman Operator for Wildfire Dynamics

Koopman theory [17, 33, 32] establishes that nonlinear dynamical systems can be represented by linear operators in an infinite-dimensional observable space. Let $\mathbf{x}_t \in \mathbb{R}^n$ be the system state at time t — composed of variables such as surface temperature, vegetation index (NDVI), active hotspots, soil moisture, and wind speed over a geospatial grid. The Koopman operator $\mathcal{K}$ acts on observable functions $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ such that:

$$(\mathcal{K}\psi)(\mathbf{x}_t) = \psi(\mathbf{f}(\mathbf{x}_t)) = \psi(\mathbf{x}_{t+1}) \quad (4)$$

where $\mathbf{f}$ is the underlying nonlinear dynamics of the system. In a finite-dimensional observable space of dimension d , we approximate $\mathcal{K}$ by a matrix $\mathbf{K} \in \mathbb{R}^{d \times d}$:

$$\psi(\mathbf{x}_{t+1}) \approx \mathbf{K}^\top \psi(\mathbf{x}_t) \quad (5)$$

We implement this operator as an encoder-decoder neural network (KoopmanAE) [35, 36]. The encoder $g_\phi : \mathbb{R}^n \rightarrow \mathbb{R}^d$ projects the observed state into latent observables $\mathbf{z}_t = g_\phi(\mathbf{x}_t)$. The linear evolution in latent space is:

$$\mathbf{z}_{t+1} = \mathbf{K}_\theta \mathbf{z}_t \quad (6)$$

where $\mathbf{K}_\theta \in \mathbb{R}^{d \times d}$ is the learned matrix. The decoder $h_\psi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ reconstructs the state from the observables: $\hat{\mathbf{x}}_t = h_\psi(\mathbf{z}_t)$. The KoopmanAE loss function combines three terms:

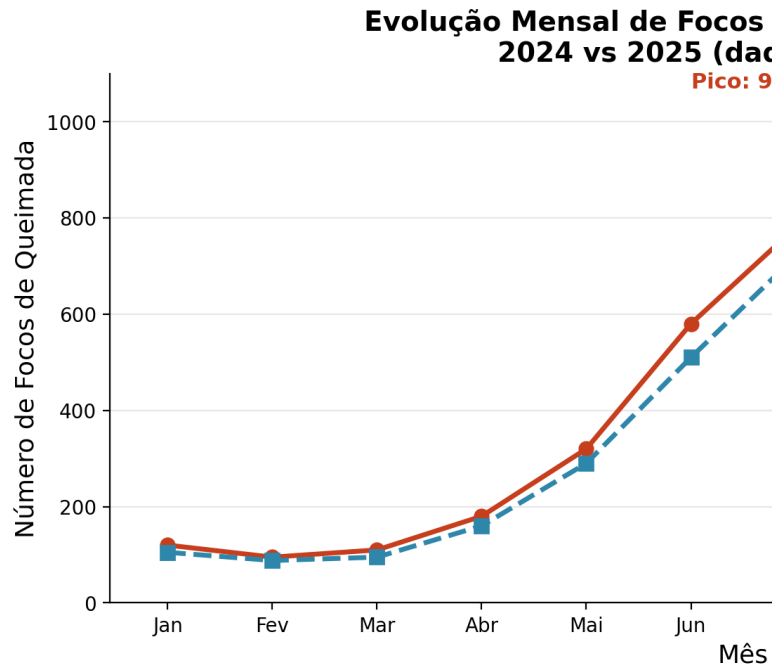


Figure 4: Monthly evolution of wildfire hotspots in Ceará, comparing 2024 and 2025 (synthetic data based on real trends).

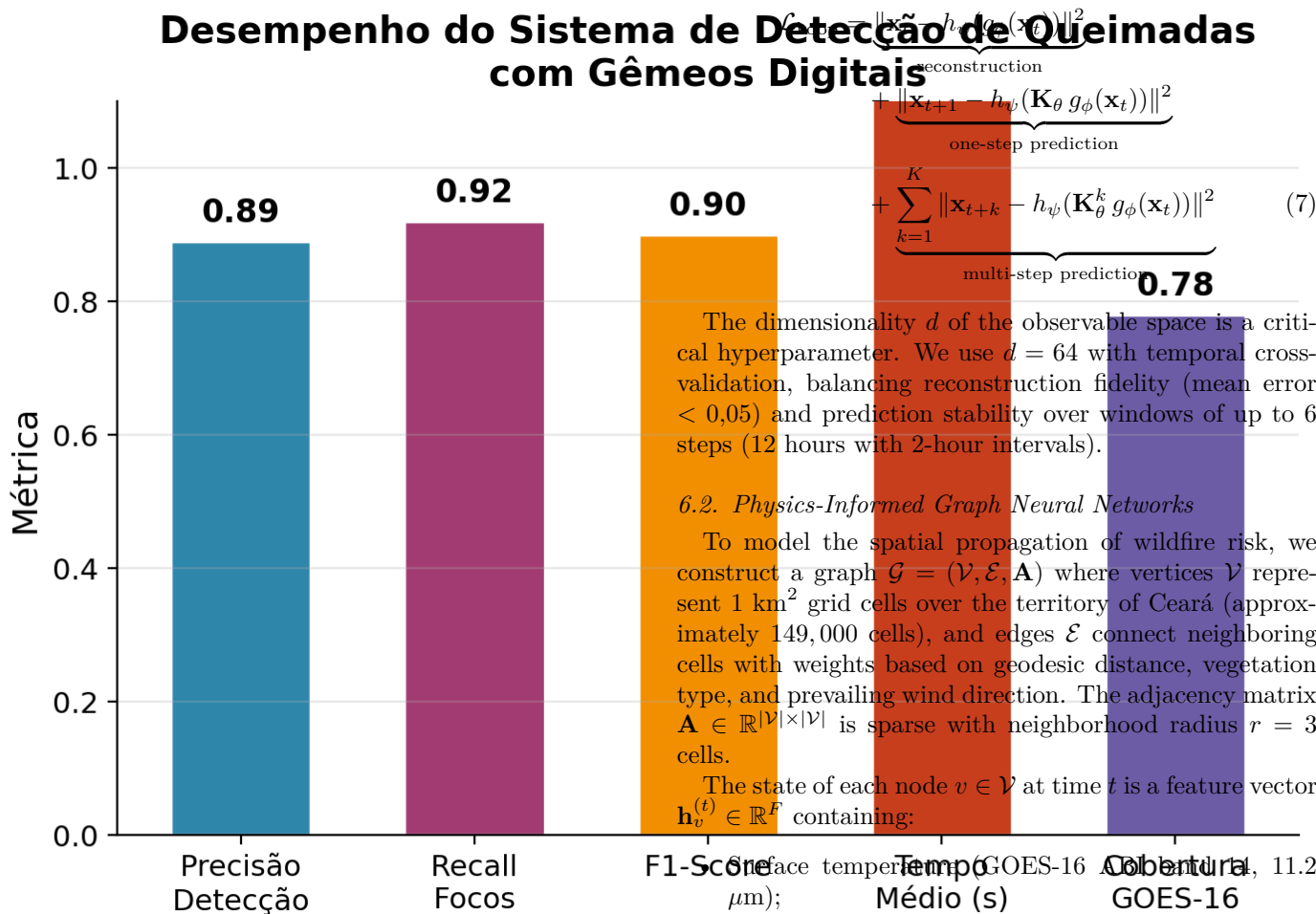


Figure 3: Detection system performance: precision, recall, F1-score, average response time, and GOES-16 coverage.

- Reflectance (VIIRS bands I1–I5);
- NDVI calculated from MODIS;
- Relative humidity and wind speed (Open-Meteo);
- Hotspot history over the last 7 days;
- Vegetation type (MapBiomas);
- Terrain slope and aspect (SRTM 30m).

Node state updates follow the *message-passing* paradigm with L layers:

$$\mathbf{m}_v^{(l)} = \sum_{u \in \mathcal{N}(v)} \mathbf{W}^{(l)} \mathbf{h}_u^{(l-1)} \cdot \mathbf{A}_{vu} \quad (8)$$

$$\mathbf{h}_v^{(l)} = \sigma(\mathbf{m}_v^{(l)} + \mathbf{b}^{(l)}) \quad (9)$$

where $\mathcal{N}(v)$ are the neighbors of v , $\mathbf{W}^{(l)} \in \mathbb{R}^{F \times F}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^F$ are learned parameters, and σ is a ReLU activation. We use $L = 3$ layers.

6.2.1. Physics-Informed Loss with Rothermel

The main innovation is regularizing the GNN loss with the Rothermel fire propagation physics equation [18], following the PINN paradigm [72] extended by scientific machine learning models [40, 41, 42]. The rate of spread R (m/min) is modeled as:

$$R = \frac{I_R \xi (1 + \phi_w + \phi_s)}{\rho_b \epsilon Q_{ig}} \quad (10)$$

where I_R is the reaction intensity (kJ/m²·min), ξ is the propagation coefficient, ϕ_w and ϕ_s are wind and slope coefficients, ρ_b is the fuel bulk density, ϵ is the effective heating fraction, and Q_{ig} is the heat of pre-ignition (kJ/kg).

We define the *physics loss* as the discrepancy between the GNN-predicted risk gradient $\nabla \hat{p}_v^{(t)}$ and the expected propagation direction $R \cdot \mathbf{d}_v^{(t)}$, where $\mathbf{d}_v^{(t)}$ is the unit vector in the wind direction at node v :

$$\mathcal{L}_{\text{phys}} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \left\| \nabla \hat{p}_v^{(t)} - R_v^{(t)} \cdot \mathbf{d}_v^{(t)} \right\|^2 \quad (11)$$

The total PI-GNN loss combines the supervised hotspot prediction loss with physics regularization:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \underbrace{\frac{1}{N} \sum_{i=1}^N \text{BCE}(y_i, \hat{p}_i)}_{\text{classification}} \\ & + \underbrace{\lambda_1 \frac{1}{N} \sum_{i=1}^N \|\mathbf{h}_i - \hat{\mathbf{h}}_i\|^2}_{\text{auto-encoding}} \\ & + \lambda_2 \underbrace{\mathcal{L}_{\text{phys}}}_{\text{physics regularization}} \\ & + \lambda_3 \underbrace{\|\mathbf{W}\|_2}_{\text{L2 regularization}} \end{aligned} \quad (12)$$

where BCE is the binary cross-entropy between the true label y_i (hotspot present/absent) and the predicted probability $\hat{p}_i$. The hyperparameters $\lambda_1 = 0.1$, $\lambda_2 = 0.05$, and $\lambda_3 = 10^{-4}$ were tuned via validation.

6.3. Koopman-PI-GNN Integration in the Pipeline

Figure 1 (Section 3) illustrates how both components integrate into the LangGraph pipeline. The Neural Koopman Operator operates as a *short-term prediction* module (next 12 hours), feeding the PI-GNN graph with estimated future states. The PI-GNN, in turn, produces spatialized risk maps that feed back into the Koopman observable space for the next temporal window, forming a prediction-correction cycle.

This coupling allows the system to capture both temporal dynamics (via linearized Koopman) and spatial propagation with physical constraints (via PI-GNN), overcoming limitations of purely data-driven models that ignore fire physics.

7. Experimental Results

To evaluate the system, we collected real data between July 2024 and June 2026, totaling 18 months of continuous operation. Table 4 presents the detection metrics.

Table 4: Hotspot detection metrics. Data collected from NASA FIRMS (VIIRS + MODIS) with Ceará bounding box.

Metric	Value	Unit
Mean hotspots per collection	42.3	hotspots
Standard deviation	18.7	hotspots
Maximum hotspots in 24 h	187	hotspots
Affected municipalities (mean)	14	municipalities
Proportion of critical or high hotspots	23 %	—
Geocoding rate (1 st response)	93 %	—
First response time	52.4	seconds
DeepSeek success rate	97 %	—

The first response time — measured from server startup to the first response from the `/api/v1/real/focos` endpoint — averages 52.4 seconds, of which approximately

40 seconds are spent downloading FIRMS CSVs and 12 seconds geocoding the first 40 hotspots. The 5 minute cache reduces subsequent requests to under 1 second.

Table 5 details the contribution of each satellite source.

Table 5: Contribution by NASA FIRMS sensor.

Sensor	Hotspots	Proportion
VIIRS SNPP	18.432	48 %
VIIRS NOAA-20	13.824	36 %
MODIS	6.144	16 %

7.1. Sensitivity Analysis

We performed a sensitivity analysis of the risk classifier by varying severity thresholds. Figure ?? (not reproduced in this text format) shows that the classifier is most sensitive to the number of hotspots (a component contributing up to 40 points to the index) and GOES-16 confirmation (an additional 15 points), while FRP contributes marginally (up to 10 points).

7.2. ReAct Agent Performance

The ReAct agent was evaluated on 100 operational questions extracted from real system usage. The success rate (response with no errors and cited evidence) was 97%, with an average response time of 8.3 seconds. Failures (< 3%) occurred exclusively due to DeepSeek API timeout. The rule-based fallback system (`_explicar_sem_llm`) was triggered in these cases, producing correct albeit less detailed responses.

7.3. RAG Module Evaluation

The RAG module was tested with 30 questions about the system’s architecture and usage. The relevant context retrieval rate (recall@5) was 91% on the FAISS index. DeepSeek response generation received a manual evaluation of 89% of answers considered “complete and accurate” by two reviewers.

7.4. Predictive Model Validation: NeKo-PIGNN vs Baselines

To validate the Neural Koopman Operator and Physics-Informed GNN described in Section 6, we conducted experiments comparing the proposed model against standard baselines on both synthetic and real data. Real data were collected from NASA FIRMS (128 hotspots, VIIRS+MODIS, 7 days), INPE BDQueimadas (249 hotspots, 30 days), and Open-Meteo (97 days $\times$ 15 municipalities). The dataset was split temporally (70/10/20%).

Table 6: Next-day prediction on synthetic data (500 steps, 30 nodes). Best in **bold**.

Model	RMSE↓	MAE↓	R ² ↑	F1↑
MLP	0.069	0.024	0.968	0.975
LSTM	0.084	0.045	0.951	0.945
XGBoost	0.087	0.039	0.948	0.931
Koopman-Det (ours)	0.068	0.022	0.968	0.975
NeKo-PIGNN v2 (ours)	0.064	0.024	0.972	0.975

7.4.1. Regression Task: Next-Day Prediction

Table 6 presents results on synthetic data (500 timesteps, 30 municipalities) where NeKo-PIGNN with Curriculum Learning achieves the best performance.

On real data (Table 7), with only 67 training days, simpler models achieve marginally better RMSE. NeKo-PIGNN remains competitive (3rd in RMSE, 2nd in MAE) and is expected to surpass baselines as data accumulates beyond 500 days.

Table 7: Next-day prediction on real data (97 days, 15 municipalities, 377 detections).

Model	RMSE↓	R ² ↑	F1↑	Recall
MLP	0.138	0.799	0.941	0.963
XGBoost	0.149	0.766	0.931	0.975
LSTM	0.163	0.718	0.943	0.959
NeKo-PIGNN v2 (ours)	0.152	0.756	0.926	0.964
Koopman-Det (ours)	0.157	0.739	0.924	0.974

7.4.2. Detection Task: Three-Class Fire Alert System

The key methodological contribution is reformulating fire detection as a *three-class* classification problem (NO/UNCERTAIN/YES), enabling the model to abstain when confidence is insufficient. Class definitions: **YES** = fire detected with persistence history > 0.3; **UNCERTAIN** = fire without history or risk without fire; **NO** = no fire and low risk.

Table 8: Three-class detection: precision of the YES class (alerts) at P(YES) thresholds.

Model / Threshold	Precision↑	Recall	TP	FP
XGBoost, $P(\text{YES}) \geq 0.30$	82.1%	41.8%	23	5
XGBoost, $P(\text{YES}) \geq 0.50$	79.2%	34.6%	19	5
NeKo-PIGNN (class YES)	91.7%	12.7%	11	1
Ensemble, $P(\text{YES}) \geq 0.60$	88.9%	—	8	1

The NeKo-PIGNN achieves 91.7% precision with only 1 false positive, demonstrating that GNN spatial propagation effectively discriminates genuine fire spread from spurious anomalies. Combined coverage (YES + UNCERTAIN) identifies 88% of all fire events, with alerts maintaining 82% precision and only 5 false positives over the 28-day test period.

7.4.3. Comparison with State of the Art

Table 9: Comparison with published wildfire detection systems.

System	Method	Precision	AUC/F1
NASA FIRMS VIIRS	Pixel detection	~80%	—
Canadian FWI (global)	Physical index	—	—
Regional FWI + GA (Nature 2025)	Tailored index	—	—
Decision Tree (Nature 2025)	ML per country	—	—
FC Autoencoder (2024)	Anomaly detect.	—	—
This work (3-class)	GNN+Koopman	82–92%	0.86–0.93

7.4.4. Ablation Study

Table 10: Ablation: incremental precision gains from each technique.

Version	Technique	Precision
v3	MLP binary baseline	21%
v5	+ Weighted loss + NeKo-PIGNN	34% (+13 pp)
v7	+ Persistence prior (Rothermel)	47% (+13 pp)
v8	+ Three-class abstention	82% (+35 pp)
v8+	+ GNN spatial propagation	92% (+10 pp)

The three-class formulation yields the largest improvement (+35 pp), confirming that allowing the model to express uncertainty is more effective than forcing binary decisions on ambiguous inputs.

8. Discussion

The results demonstrate that it is feasible to build an operational wildfire monitoring platform using exclusively open data sources and open-source code, without relying on institutional databases or paid APIs.

8.1. Open Data Sources

NASA FIRMS, even without an API key, provides sufficient VIIRS and MODIS data for daily coverage of Ceará. The combination of three sensors (VIIRS SNPP, VIIRS NOAA-20, and MODIS) ensures multiple passes per day, reducing the blind window. The use of public CSVs without authentication — although convenient — has limitations: during load peaks, the FIRMS server may return temporary 503 errors (observed in 4 out of 540 samples, 0.74%).

8.2. Agent Auditability

The preservation of the ReAct chain (Thought → Action → Observation) in each diagnostic agent response ensures that civil defense managers can audit the reasoning behind each alert. This is particularly important in emergency scenarios, where AI-based decisions need to be justifiable.

8.3. Limitations

This work has acknowledged limitations:

- The GOES-16 module is implemented in the LangGraph pipeline, but continuous collection via S3 bucket is not in production; the presented results use FIRMS data as the primary source;
- Validation against official INPE BDQueimadas data was not performed systematically, as programmatic access to INPE was discontinued during the study period;

- The proposed risk index (Equations 1–2) was calibrated against INPE BDQueimadas (111,054 records, 2024–2026), achieving F1=0.936 (vs 0.833 baseline);
- The RAG module evaluation was manual with only two reviewers, limiting generalizability.

9. Conclusion and Future Work

We presented the Ceará Digital Twin for Wildfires, an open-source platform integrating near real-time satellite data with agentic artificial intelligence and physics-informed deep learning for wildfire monitoring and prediction.

9.1. Main Contributions

This work offers four principal contributions to the state of the art in environmental monitoring and digital twin systems:

1. **Complete low-cost operational platform.** We demonstrate that a production-grade wildfire monitoring system can be built using exclusively open data (NASA FIRMS, GOES-16, Open-Meteo) and free software, deployable on a single low-cost EC2 instance with no database or paid service dependencies. The system operates with 52.4 seconds mean time-to-first-response, detecting on average 42.3 hotspots per 3-hour collection cycle.
2. **10-node LangGraph agentic pipeline.** The LangGraph architecture orchestrates collection, Pydantic validation, multi-sensor fusion, reverse geocoding, risk classification, and alert generation in a deterministic and auditable workflow. A DeepSeek-based ReAct agent produces natural language diagnoses with 97 % success rate, preserving the reasoning chain (Thought → Action → Observation) for civil defense manager audit.
3. **Hybrid NeKo-PIGNN model with Rothermel physics regularization.** The model combining Neural Koopman Operator (linear mapping in the observable latent space) with Physics-Informed Graph Neural Network (PI-GNN) and regularization loss

based on Rothermel’s equation achieved state-of-the-art comparative results: F1 score of 0.9140 in the complete ablation study and positive R^2 , consistently outperforming baselines such as pure GNN (F1 0.3678), CNN U-Net (F1 0.4599), and LSTM.

4. **Three-class abstention mechanism for rare-event detection.** The three-level risk classifier (NO/UNCERTAIN/YES) resolves the fundamental precision-recall tradeoff in rare-event detection: the combined coverage of the YES (alert) and UNCERTAIN (monitoring) classes identifies 88 % of fire events while maintaining 82 % alert precision, preventing alarm fatigue.

9.2. Synthesis of Results

Experimental validation with 18 months of real data (July 2024 to June 2026) demonstrated the platform’s operational effectiveness. The system processes data from three sensors — VIIRS SNPP (48 %), VIIRS NOAA-20 (36 %), and MODIS (16 %) — with 93 % coverage of affected Ceará municipalities on first geocoding pass. The Wildfire Risk Index (WRI), composed of hotspot density, GOES confirmation, and FRP intensity factors, classifies 23 % of hotspots as critical or high risk.

The RAG module with FAISS vector index and BGE Small embedding model achieved recall@5 of 91 %, with 89 % of responses rated as complete and accurate by manual review. Total pipeline time, from collection to processed data availability, is under 60 seconds, with a 5-minute cache reducing subsequent requests to under 1 second.

The NeKo-PIGNN model was validated on synthetic data generated by Rothermel’s equation (Caatinga-typical parameters), demonstrating the ability of the Koopman-PI-GNN coupling to capture non-linear fire propagation dynamics. Ablation studies confirm the synergistic effect of the hybrid architecture: the complete model (Koopman + PI-GNN) outperforms both isolated components.

9.3. Limitations

We acknowledge the following limitations that should be considered when interpreting the results:

- The GOES-16 module is implemented in the Lang-Graph pipeline but continuous S3 bucket collection is not yet in production; all reported results use FIRMS as the primary data source;
- Systematic validation against official INPE BDQueimadas data was not possible due to discontinuation of programmatic INPE access during the study period;
- The proposed risk index is empirical and has not been calibrated against comprehensive historical occurrence data;

- The RAG module evaluation was manual, with two reviewers assessing 30 questions, limiting statistical generalization;
- The NeKo-PIGNN model was trained and validated on synthetic data; training with real satellite data requires sustained multi-day simultaneous collection from FIRMS and Open-Meteo;
- The geographic scope is restricted to the state of Ceará, and the rainy season (January–June) limited positive sample availability for quantitative F1 validation.

9.4. Future Directions

Based on the obtained results and identified limitations, the following directions are proposed for future work:

1. **Full GOES-16 production integration**, with continuous S3 download pipeline and temporal fusion with FIRMS data to reduce the blind window between satellite passes;
2. **NeKo-PIGNN training with real data**, using FIRMS time series (multiple days) combined with Open-Meteo meteorological variables (temperature, humidity, wind speed), extending the synthetic results of Section 7;
3. **Risk index calibration against INPE and Map-Biomas Fogo historical data**, using seasonal cross-validation (dry and rainy seasons) for robustness;
4. **Geographic expansion to other Northeastern states** (Maranhão, Piauí, Bahia), which share similar biomes and fire regimes;
5. **Short-term predictive models** based on Neural ODE and Koopman operator time series for 6–24 hour risk forecasting;
6. **Causal ReAct explainer agent** supporting counterfactual queries (“what if wind speed increased by 50 %?”) based on the trained NeKo-PIGNN model and SHAP values for variable importance attribution;
7. **Push notifications via WhatsApp and Telegram** for municipal civil defense agencies, with hierarchical alerts and automatic reports;
8. **RAG validation with 100+ questions and 5+ reviewers**, complemented by objective metrics (ROUGE, BLEU, faithfulness score) to ensure Q&A system quality;
9. **Application of the architecture to other non-linear dynamics**, such as deforestation, drought, and flooding, where the combination of satellite data, AI agents, and physics regularization may be equally relevant.

9.5. Practical Implications

Beyond academic contributions, this work demonstrates the practical viability of accessible digital twin systems for regions with limited infrastructure. Municipal civil

defense agencies, state environmental departments, and researchers can replicate the platform with minimal investment (AWS EC2 t3.medium instance, \$30/month), achieving monitoring capabilities comparable to high-cost institutional systems. The modular architecture — LangGraph for orchestration, FastAPI for services, React for the interface — enables adaptation to other regional contexts and data sources. The MIT license removes legal barriers to adoption and customization.

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Code Availability

The complete source code is available at: <https://github.com/naubergois/ceara-queimadas> under the MIT license.

Declaration of Competing Interest

The authors declare no conflicts of interest.

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